1. Clinical Question & Core Hypothesis

Clinical Question

NMIBC patients often show organized local immune structures (TLS/TILs) after BCG yet some fail to achieve durable clinical benefit — why does local immune organization not always translate into therapy response? 2) Tier■1 basis: I built on Core Path 1 (BCG → biomarker loop), Core Path 2 (PD■L1 ↔ bTMB / immune checkpoint dynamics), and Core Path 3 (MSH6 as a mismatch■repair/prognostic axis). These provide a mechanistic chain: mismatch■repair status (MSH6) alters mutation/neoantigen load, which modulates PD■L1–mediated adaptive resistance and thereby the functional output of TLS. 3) Tier■2 support: noninvasive urine DNA methylation panels and cross■cancer methylation markers (SDC2/TFPI2) provide feasible stratification/readout routes; TP53/ER axis and pZVR supply orthogonal prognostic signals that can validate or refine the mechanistic subgrouping. 4) Endpoint choice rationale: I selected therapy_response (BCG response proxy) because the core question asks why an immune therapy (BCG) sometimes fails despite apparent local immune engagement — a therapy■response endpoint directly tests that mechanism. 5) Deliberate exclusions: I limited the hypothesis to one imaging feature (TLS) and two anchor genes (MSH6, PD■L1) to meet the focusing rule; I excluded explicit mention of bTMB and methylation markers from the hypothesis sentence (they are in secondary_modifiers) to avoid overloading the mechanistic claim and to keep the question sharply testable?

Working Hypothesis

TLS-like lymphoid aggregates in NMIBC predict a reduced response to intravesical BCG in tumors that harbor MSH6 loss due to increased neoantigen burden driving PD-L1 upregulation and adaptive immune resistance within TLS; consequently, TLS-mediated anti-tumor activity is blunted when MSH6 is deficient. Clinically, TLS-rich NMIBC with MSH6 deficiency and high PD-L1 expression will show lower therapy_response to BCG, and these patients may benefit from PD-1/PD-L1 blockade to restore TLS-driven immunity and improve clinical outcomes.

Element	Value
Imaging feature	TLS-like lymphoid aggregates
Anchor gene(s)	MSH6, PD-L1
Primary endpoint	therapy_response

2. Why This Gene — Not Others

Primary hypothesis gene(s)

MSH6: MSH6, PD■L1) to meet the focusing rule; I excluded explicit mention of bTMB and methylation markers from the hypothesis sentence (they are in secondary_modifiers) to avoid overloading the mechanistic claim and to keep the question sharply testable.

PD-L1: Selected as a core mechanistic driver from Tier 1 knowledge graph analysis.

Alternative / secondary genes considered

Tier	Gene	Role / Reason excluded from core

Secondary	bTMB	Supporting modifier from knowledge graph
Secondary	urine DNA methylation biomarkers	Supporting modifier from knowledge graph
Secondary	SDC2	Supporting modifier from knowledge graph
Secondary	TFPI2	Supporting modifier from knowledge graph
Secondary	TP53	Supporting modifier from knowledge graph
Secondary	ER_status	Supporting modifier from knowledge graph
Secondary	pZVR	Supporting modifier from knowledge graph

3. Clinically Testable Predictions

Prediction 1: Retrospective cohort analysis of NMIBC patients treated with BCG with data on TLS density/maturation, MSH6 status, PD-L1 expression, and bTMB;

Prediction 2: prospective collection of tumor biopsies for IHC (TLS, PD-L1), targeted sequencing for MSH6, calculation of bTMB, and outcome data on therapy_response;

Prediction 3: in vitro/in vivo models to test whether MSH6 loss leads to PD-L1 upregulation and TLS dysfunction, and whether PD-1/PD-L1 blockade can restore TLS activity and improve BCG-like responses.

4. Minimum Viable Validation Plan

A. Retrospective Cohort Analysis

Data source: TCGA-BLCA (Firehose Legacy), ~400 patients - Cox PH model with interaction term for the imaging feature and anchor gene - Kaplan-Meier 2x2 stratification

B. Tissue-Level Validation

Methods: Multiplex IF / IHC / spatial transcriptomics

Key readouts: - Immune structure markers (e.g., CD20, CD3, PNAd) - Effector cell localization (CD8) relative to tumor nests - Anchor gene protein / phospho-protein expression - Stromal features (collagen density by SHG, if applicable)

Goal: Confirm that the proposed mechanism operates at the spatial / tissue level.

C. Functional Validation

Models: Ex vivo organotypic cultures, 3D co-culture, syngeneic mouse models

Perturbations: Genetic (KO/rescue) or pharmacologic modulation of anchor gene

Readouts: T-cell infiltration, cytotoxicity (granzyme B / IFN-gamma), tumor growth / survival

Goal: Establish causal relationship between anchor gene activity and immune efficacy.

5. Strengths, Caveats & Alternative Directions

Strengths

- Novel mechanistic hypothesis derived from knowledge graph analysis.

Caveats & Risks

- Gene expression does not equal protein activity; phospho-protein or functional assays are needed for confirmation.

Alternative Directions

- bTMB: Could serve as an alternative mechanistic axis if the primary hypothesis is not supported.
- urine DNA methylation biomarkers: Could serve as an alternative mechanistic axis if the primary hypothesis is not supported.